

Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	1 July 2026
Team ID	
Project Name	
Maximum Marks	4 Marks

Functional Requirements – HouseRent (MERN Stack House Rental Platform)

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Registration	Registration through FormOwner RegistrationRole Selection (User/Owner)
FR-2	User Authentication	Login using Email & PasswordJWT AuthenticationLogout
FR-3	Password Management	Forgot Password via EmailReset Password using Secure Link
FR-4	Property Browsing	View Available PropertiesSearch PropertiesView Property Details
FR-5	Property Booking	Book Rental PropertyView Booking HistoryCancel Booking (Optional)
FR-6	Property Management (Owner)	Add PropertyEdit PropertyDelete PropertyUpload Multiple Images
FR-7	Owner Approval	Admin Approves or Rejects Owner Registration
FR-8	Admin Dashboard	Manage UsersManage PropertiesManage BookingsMonitor Platform Activities
FR-9	Profile Management	View ProfileEdit Profile
FR-10	Image Upload	Upload Multiple Property Images using Cloudinary

Non-Functional Requirements – HouseRent (MERN Stack House Rental Platform)

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The application should provide a simple, responsive, and user-friendly interface that is easy to navigate on desktop and mobile devices.
NFR-2	Security	The system must use JWT authentication, password encryption (Bcrypt), protected routes, and role-based access control to secure user data.
NFR-3	Reliability	The application should store data accurately, handle errors gracefully, and maintain data consistency during all operations.
NFR-4	Performance	Pages should load quickly, API responses should be fast, and image uploads should be optimized for better performance.
NFR-5	Availability	The platform should be available to users at all times with minimal downtime and reliable server connectivity.
NFR-6	Scalability	The system should support increasing numbers of users, properties, and bookings, and allow easy integration of future features like payments, reviews, and notifications.